

EXPT NO:4	TRANSFORM DATA TO SUIT BI REPORTING REQUIREMENTS AND DESIGN A DASHBOARD WITH KEY PERFORMANCE INDICATORS (KPIs) USING TABLEAU AND POWERBI
DATE: 24.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is the role of data transformation in BI reporting?

Data transformation converts raw data into a clean and structured format suitable for analysis. It improves data quality by correcting inconsistencies, handling missing values, and creating calculated fields, enabling accurate and meaningful BI reports.

2. What are the common data cleaning techniques in Tableau and Power BI?

Common data cleaning techniques include:

- Removing duplicate records
- Handling missing or null values
- Correcting data types (such as converting text to dates or numbers)
- Renaming columns
- Filtering unnecessary data
- Creating calculated fields for analysis

These steps improve the accuracy and reliability of dashboard visualizations.

3. How do Tableau and Power BI handle data aggregation?

Tableau and Power BI aggregate data using functions such as SUM, AVERAGE, COUNT, MIN, and MAX. Aggregation helps summarize large datasets and enables meaningful comparisons across countries, regions, or time periods.

4. What are KPIs, and why are they important in a dashboard?

Key Performance Indicators (KPIs) are measurable values that indicate the performance of a process or objective. In dashboards, KPIs provide quick insights into important metrics such as total cases, total deaths, vaccination coverage, and case fatality rate, helping users make informed decisions.

5. What are the different types of charts available in Tableau and Power BI for visualization?

Both Tableau and Power BI support various chart types, including:

- Line Charts
- Bar Charts
- Column Charts
- Pie Charts
- Maps
- Scatter Plots
- Area Charts
- KPI Cards
- Tables and Matrices

IN-LAB EXERCISE:

OBJECTIVE:

To transform COVID-19 data into a structured format suitable for Business Intelligence (BI) reporting and design interactive dashboards with Key Performance Indicators (KPIs) using Tableau and Power BI.

PROCEDURE:

1. Data Collection:

- Import the **COVID-19 dataset** from inbuilt sources like *Our World in Data* or CSV files containing country-level COVID statistics (e.g., new cases, deaths, vaccinations).
- Datasets may include fields such as *date*, *location*, *new_cases*, *total_cases*, *new_deaths*, *total_deaths*, *people_vaccinated*, etc.

2. Data Cleaning and Transformation:

- Convert columns like *date* to proper date formats.
- Handle missing or null values in *new_cases*, *total_cases*, etc.
- Create calculated fields (e.g., daily case growth rate, case fatality rate, vaccination percentage).

3. Data Aggregation:

- Aggregate data by *continent*, *country*, or *month*.
- Use measures like SUM(new_cases), AVG(positivity_rate), MAX(total_deaths).

4. Designing KPIs:

- Identify KPIs such as:
 - Total Confirmed Cases
 - Total Deaths
 - Case Fatality Rate
 - Vaccination Coverage (%)
 - Daily Case Growth Rate
- Use DAX (Power BI) or Calculated Fields (Tableau) to compute and visualize these KPIs.

5. Dashboard Development:

- Build visualizations such as:
 - Line charts for daily new cases and deaths
 - Bar charts for country-wise vaccination progress
 - Maps to show regional case distribution

- KPI cards for high-level insights
- Use filters (e.g., by continent, country) and slicers (e.g., by date range) for interactivity.

6. Publishing and Sharing:

- Publish the dashboard using Tableau Public/Server or Power BI Service.
- Share the dashboards with stakeholders through public or organizational portals.

Experiment Dataset: COVID-19

Global Dataset

Fields Used:

- **Date** – Date of the report
- **Location** – Country or region
- **New Cases** – Daily reported cases
- **New Deaths** – Daily reported deaths
- **People Vaccinated** – Cumulative vaccinated count
- **Total Cases** – Cumulative confirmed cases
- **Total Deaths** – Cumulative deaths
- **Population** – Total population of the country

OUTPUT :

Power BI :

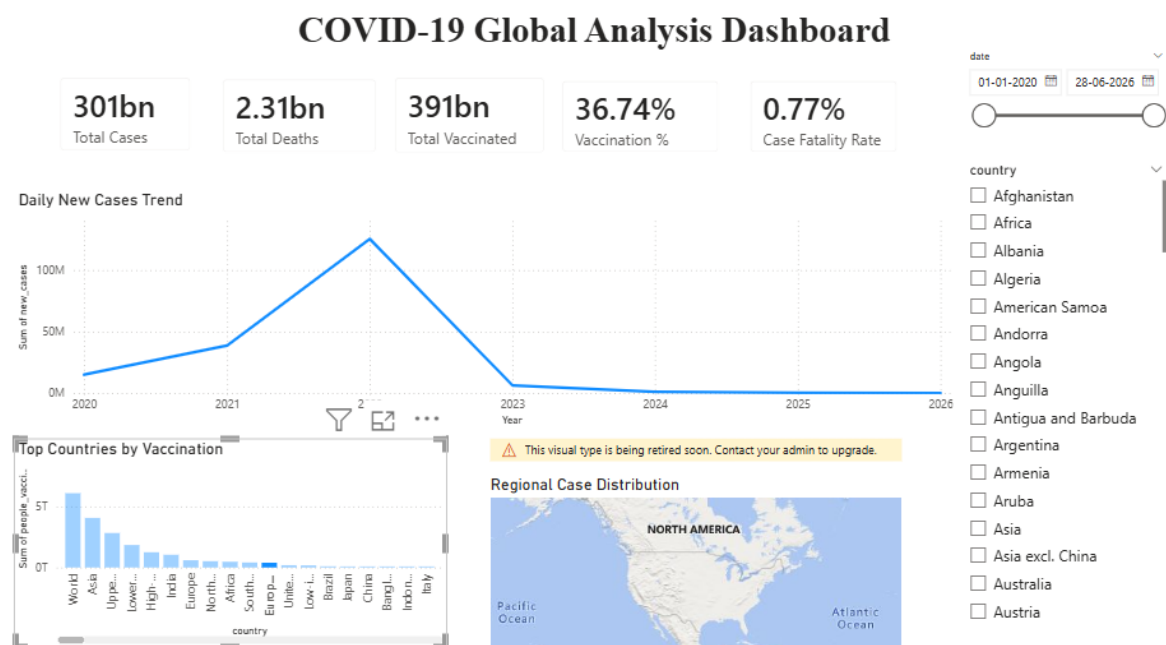
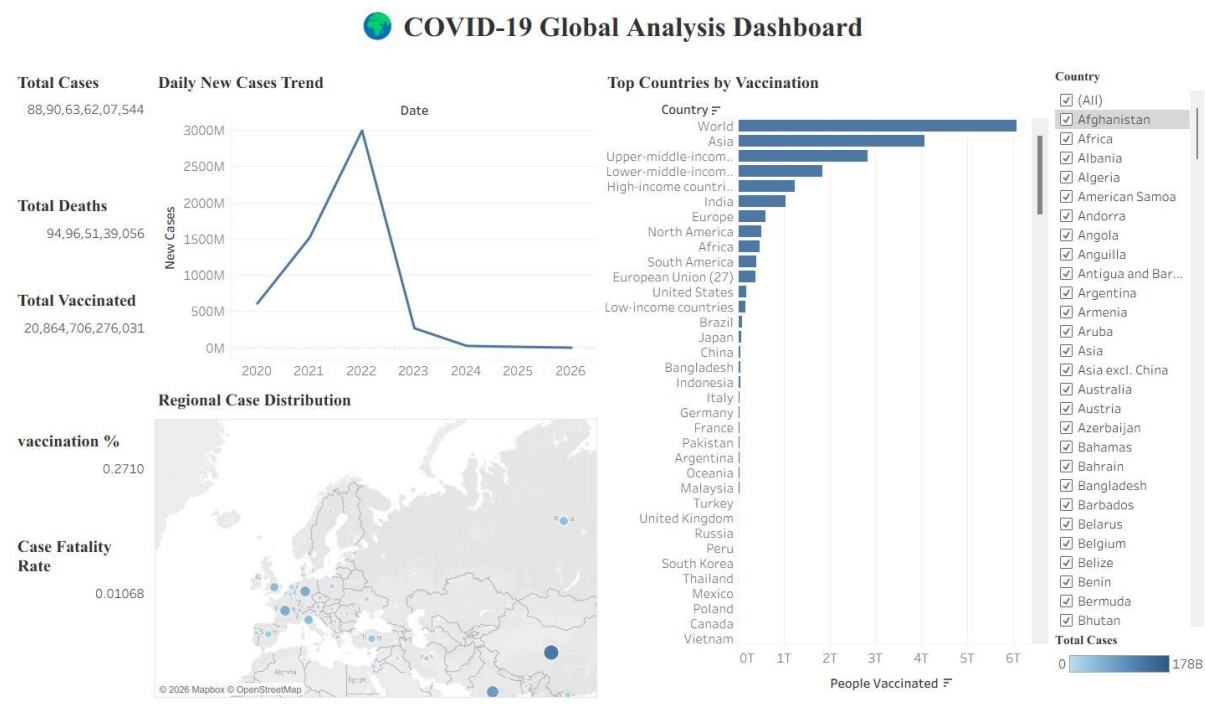


Tableau :



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did data transformation improve the quality of insights in your COVID-19 dashboard?

Data transformation improved the quality of the dashboard by converting the date column into a proper date format, handling missing values in fields such as people_vaccinated, and ensuring numeric fields like total_cases, total_deaths, and population were in the correct format. It also enabled the creation of calculated KPIs such as Vaccination Percentage and Case Fatality Rate, making the dashboard more accurate and meaningful for analysing COVID-19 trends.

2. What challenges did you encounter during the data cleaning process, and how did you resolve them?

The main challenges were null values in the people_vaccinated column and incorrect data types for some numeric fields. These issues caused calculation errors while creating KPIs. The problems were resolved by converting the columns to the correct numeric data types, handling missing values appropriately, and validating the dataset before creating measures and visualizations.

3. Which KPI provided the most valuable insight, and why?

The Vaccination Percentage KPI provided the most valuable insight because it compares the number of people vaccinated with the total population, allowing easy comparison of vaccination coverage across countries. Along with the Case Fatality Rate, it helped evaluate both vaccination progress and the severity of COVID-19 in different regions. These KPIs were part of the dashboard requirements for the experiment.

4. How does interactivity in dashboards improve user experience?

The interactive dashboard allows users to filter data using Country and Date slicers. Users can analyse COVID-19 statistics for a specific country or time period without changing the original dataset. The line chart, bar chart, map, and KPI cards update automatically based on the selected filters, making the dashboard easier to explore and understand.

5. What steps would you take to further enhance the dashboard's effectiveness? To improve the dashboard further, I would:

- Add more KPIs such as Recovery Rate and Death Rate per Million.
- Include drill-down analysis from continent to country.
- Add trend forecasting for future COVID-19 cases.
- Display the last data refresh date.
- Use a live COVID-19 data source so that the dashboard automatically updates with new information.

RESULT:

The COVID-19 dataset was successfully cleaned, aggregated, and transformed for visual analysis using Tableau and Power BI. Missing values, data type mismatches, and inconsistencies were addressed. KPIs such as vaccination rate and daily case growth offered valuable insights into the pandemic trends across countries and timeframes. The resulting interactive dashboards allowed for effective public health monitoring and communication.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		